Ontology Engineering

A Simple Ontology Engineering Methodology

1. Determine the domain and scope of the ontology
2. Consider reusing existing ontologies
3. Enumerate terms in the ontology
4. Define the classes and the class hierarchy
5. Define the properties of classes
6. Define the characteristics of properties
7. Create individuals

Step 1: Determine the Domain and Scope

What is the domain that the ontology will cover?

Road signs as used in the European Union.

For what are we going to use the ontology?

To identify the **type of road sign** based on its properties and to determine **to which kind of agent it applies** to in order to choose the action one needs to take in traffic.

For what types of questions should the information in the ontology provide answers (competency questions)?

What are the required background and border colours for a model B2a "STOP" sign?

Who will use and maintain the ontology?

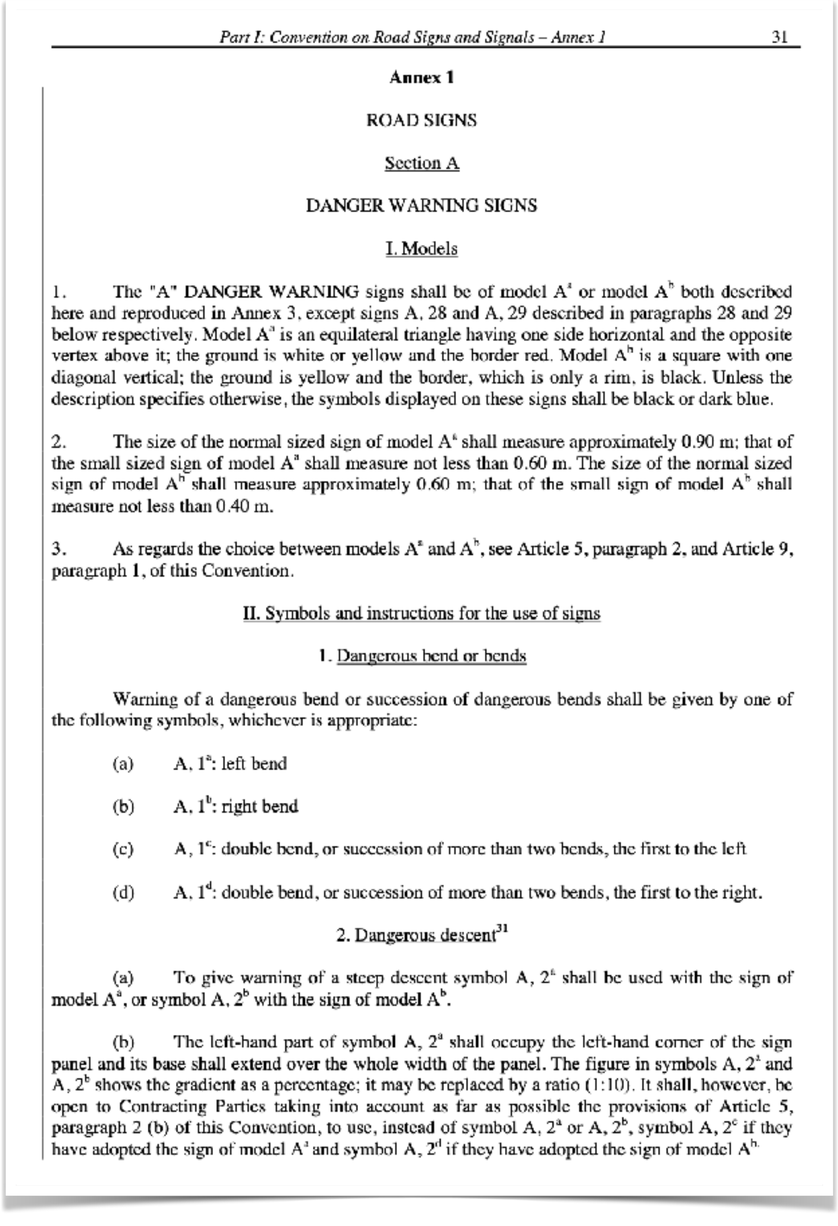
For example, (semi-)autonomous vehicles and their engineers.

Competency Questions

- **CQs about the taxonomy of road signs**
  - What are the primary kinds of road signs distinguished in the Vienna Convention?
  - Is a priority sign considered a type of road sign?
- **CQs about road sign properties**
  - Which border colours can circle-shaped road signs have?
  - What are the required background and border colours for a model B2a "STOP" sign?
  - What is the standard measurement for the side of a normal-sized danger warning sign?
- **CQs about the pragmatics of road signs**
  - Which danger warning symbol (A11a or A11b) must be used to warn of falling rocks?
  - What specific symbol is used on sign A15 to warn of "Cattle or other animals crossing”?
  - Which road signs are relevant to pedestrians?
- **CQs relevant to the particular dataset**
  - How many road signs are depicted in the image IMG_20230314_49094.jpg?
  - Which images contain a “COMPULSORY MINIMUM SPEED” sign?
  - What percentage of road signs depicted in the image repository are prohibitory signs?
  - Is there any road sign in the image repository that informs about a restriction applicable to lorries or other large vehicles?
  - What is the maximum speed shown on most speed limit signs?

Step 2: Consider Reusing Existing Ontologies

- **Vienna Convention on Road Signs and Signals**
  (uploaded on the Virtual Campus)
- Part I, Annex 1



Annex 1

ROAD SIGNS

Section A

DANGER WARNING SIGNS

I. Models

1. The "A" DANGER WARNING signs shall be of model A^a or model A^b both described here and reproduced in Annex 3, except signs A, 28 and A, 29 described in paragraphs 28 and 29 below respectively. Model A^a is an equilateral triangle having one side horizontal and the opposite vertex above it; the ground is white or yellow and the border red. Model A^b is a square with one diagonal vertical; the ground is yellow and the border, which is only a rim, is black. Unless the description specifies otherwise, the symbols displayed on these signs shall be black or dark blue.

2. The size of the normal sized sign of model A^a shall measure approximately 0.90 m; that of the small sized sign of model A^a shall measure not less than 0.60 m. The size of the normal sized sign of model A^b shall measure approximately 0.60 m; that of the small sign of model A^b shall measure not less than 0.40 m.

3. As regards the choice between models A^a and A^b, see Article 5, paragraph 2, and Article 9, paragraph 1, of this Convention.

II. Symbols and instructions for the use of signs

1. Dangerous bend or bends

Warning of a dangerous bend or succession of dangerous bends shall be given by one of the following symbols, whichever is appropriate:

(a) A, 1^a: left bend

(b) A, 1^b: right bend

(c) A, 1^c: double bend, or succession of more than two bends, the first to the left

(d) A, 1^d: double bend, or succession of more than two bends, the first to the right.

2. Dangerous descent[31]

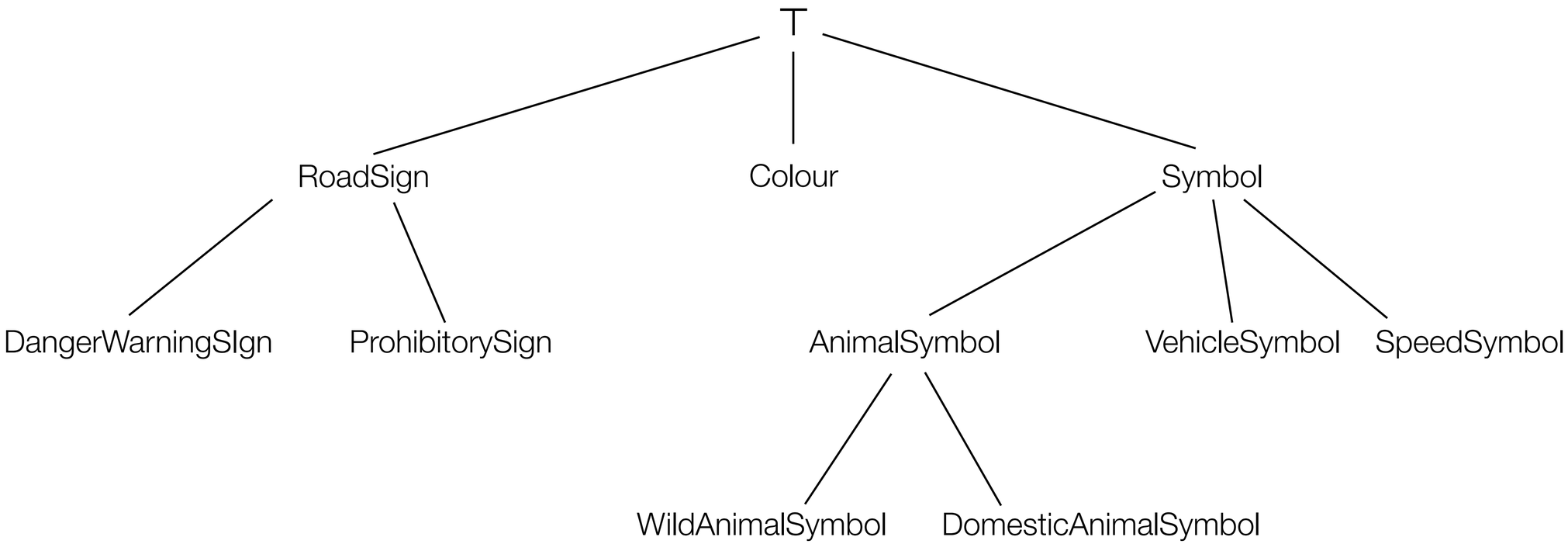
(a) To give warning of a steep descent symbol A, 2^c shall be used with the sign of model A^a, or symbol A, 2^b with the sign of model A^b.

(b) The left-hand part of symbol A, 2^a shall occupy the left-hand corner of the sign panel and its base shall extend over the whole width of the panel. The figure in symbols A, 2^a and A, 2^b shows the gradient as a percentage; it may be replaced by a ratio (1:10). It shall, however, be open to Contracting Parties taking into account as far as possible the provisions of Article 5, paragraph 2 (b) of this Convention, to use, instead of symbol A, 2^a or A, 2^b, symbol A, 2^c if they have adopted the sign of model A^a and symbol A, 2^d if they have adopted the sign of model A^b.

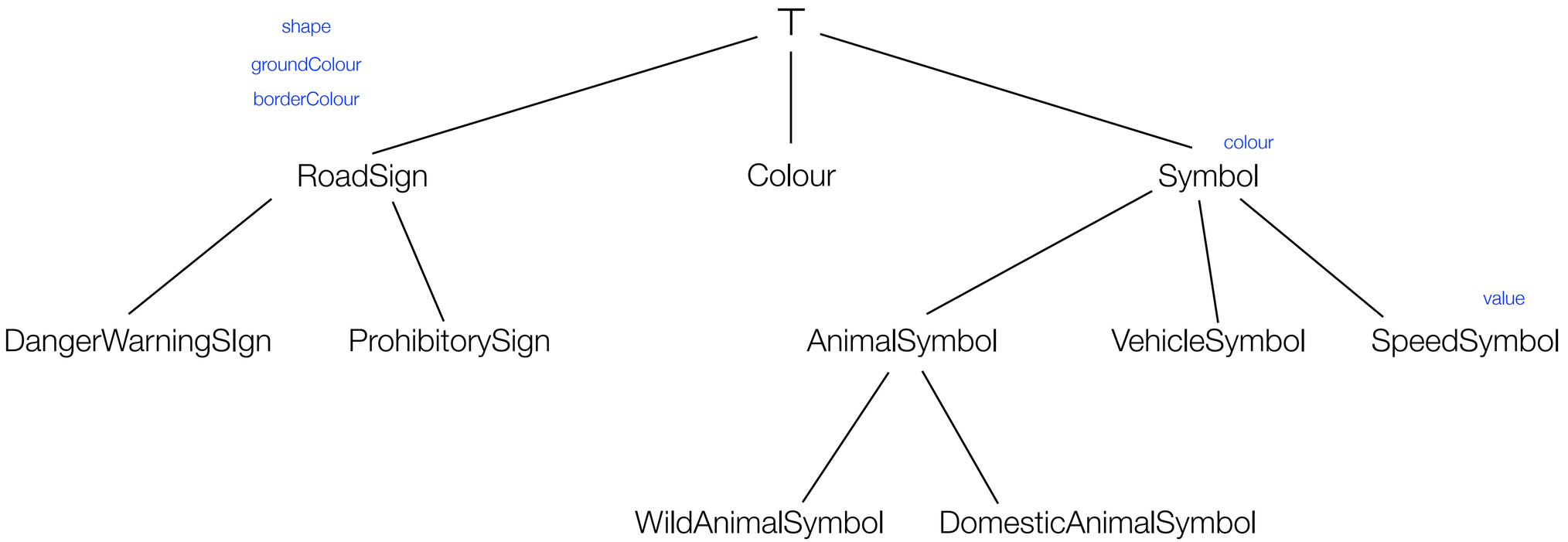
Step 3: Enumerate Important Terms

- road sign, danger warning sign, prohibitory sign, mandatory sign, …
- ground colour, border colour, shape, symbol
- white, red, blue, black, … triangle, circle, rectangle, …
- pedestrian crossing, horizontal bar, compulsory speed limit, priority road
- pedestrian, bicycle, car, lorry, motor vehicle,
- …

Step 4: Define the Classes and the Hierarchy



Step 5: Define the Properties of Classes



Step 6: Define the Characteristics of Properties

- **Value type:** determines the **range of the property**
  - *data types*: string, boolean, integer, float, enumerated, ...
  - *classes*: Colour, Shape, Symbol, ...

- C**ardinality**
  - *single cardinality* (at most one value): …
  - *multiple cardinality* (any number of values): property “symbol” of class “RoadSign”
  - *required* (at least one value)
    - required single (exactly one value): property “shape” of class “RoadSign” (functional)
    - required multiple (any number of values, but at least one): property ”symbolColour” of class “Symbol”
  - *arbitrary minimum and maximum cardinality*

Step 7: Create Instances











LLM-based Ontology Engineering

- Prompting – specify as detailed as possible the engineering step
- Revising the output
- ….

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The Prompt

Role: You are an expert Ontology Engineer specialising in [generic domain of the ontology]

Objective: [describe the task to do]

Scope & Domain:

- Domain: [define the concrete scope of the domain to be modelled]
- Key Entities: [provide key entities that need to be included in the ontology]
- Boundaries: [define what NOT to include to prevent scope creep]

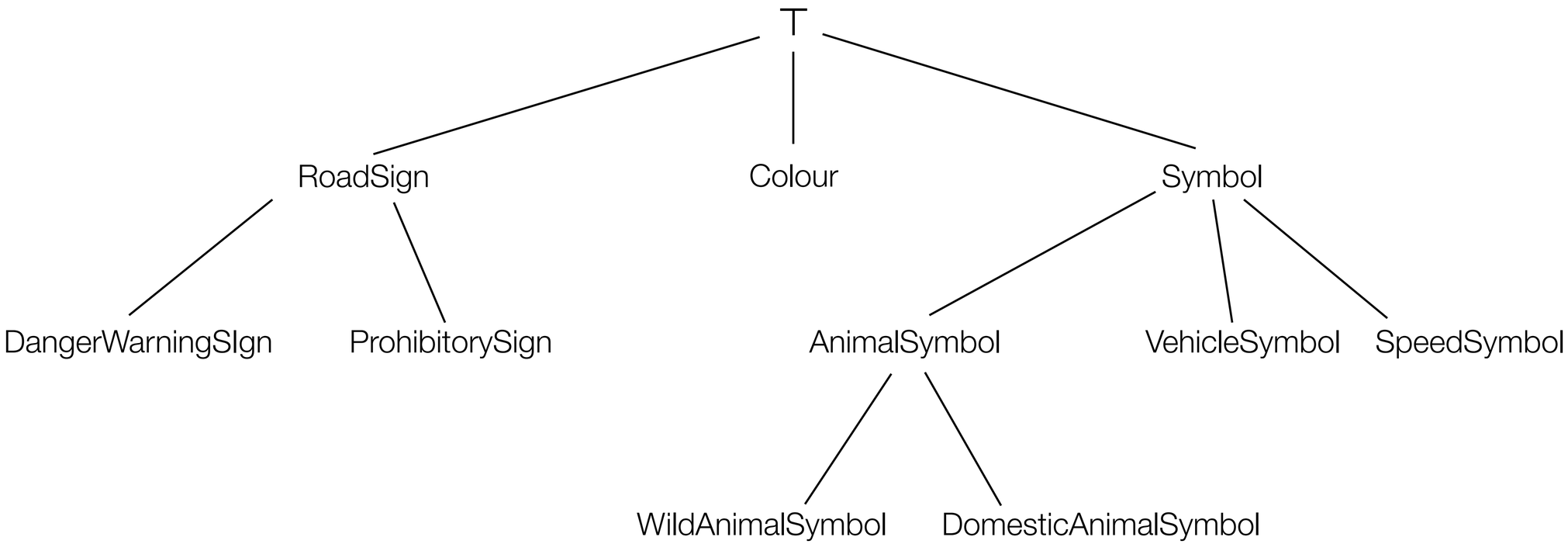
Competency Questions: The ontology must be able to answer the following questions:

- [Competency Question 1]
- [Competency Question 2]
- …

Structural Requirements: [decribe particular implementation requirements for fhis task]

Output Format: [specify the concrete output format (XML/RDF, Turtle, …)

Step 4: Define the Classes and the Hierarchy



The Prompt

Role: You are an expert Ontology Engineer specialising in the domain of traffic regulations.

Objective: Develop a formal taxonomy of road signs as an IS-A class hierarchy, written in OWL2, as the first step of an ontoloogy engineering process.

Scope & Domain:

- Domain: The taxonomy should cover all road signs and their taxonomic structure, as described in Part I, Annex 1, of the attached PDF document "Conv_road_signs_2006v_EN.pdf".
- Key Entities: The class hierarchy should be rooted in a class named *RoadSign*. Subclasses should follow the naming conventions in the attached PDF document.
- Boundaries: Only declare *RoadSign* and the subclasses mentioned in Part I, Annex 1, of the attached PDF document, but no other ontological entities. Do not generate the whole ontology in this step.

Competency Questions: When developing the taxonomy, take into account that the final ontology to be developed should be able to answer the following questions:

- [Competency Question 1]
- [Competency Question 2]
- …

Structural Requirements: Add disjointness axioms to keep classes and subclasses separate whenever this is the case.

Output Format: Generate the class hierarchy in RDF/XML sytntax.

Submission

- A link to the open data repository of the domain of your choice
    - e.g. https://analisi.transparenciacatalunya.cat/ca/Ci-ncia-i-Tecnologia/Grups-de-recerca-de-Catalunya/ufpk-rk8r/about_data
- A list of at least 20 varied competency questions (more are welcome!)
- The OWL file with the class hierarchy
- If you are following an LLM-based ontology engineering methodology, declaration of the models and main prompts used
- Submission via the Virtual Campus (https://cv.uab.cat) by 21st October at 23:59 h.